Kriyānighaṇṭu Synonym Finder

A context-aware synonym finder for Sanskrit verbs, built on a Neo4j graph database

1. Problem

A Sanskrit verb can carry several meanings, and its synonyms depend on which one is meant. गृह्णाति may mean *to seize*, *to understand* or *to be angry* depending on the passage. Given a passage (typed, a .txt file or a PDF) and one word in it, the system decides which meaning the context points to and returns the best synonym for that meaning from the Kriyānighaṇṭu dataset. When it cannot, it says why.

2. Data

Three spreadsheets were provided, with no ready-made relations between meanings:

File	Contents
Verb-Entries.xlsx	Verbs, their forms, examples, meaning labels (e.g. गतौ) and See Also references
Look-up entry.xlsx	Each meaning label with its English gloss
Root-Entry.xlsx	Verbal roots (dhātu), gaṇa and root meaning

3. Graph model

The schema follows the project's demo schema: a **Verb** reaches a **Sense** only through a **Shade**. Two verbs are synonyms in a context when they share a Shade.

```
(:Verb)-[:HAS_SHADE {id: V808.S219.1, confidence, sources}]->(:Shade)<-[:HAS_SHADE]-(:Sense)
(:Verb)-[:LISTED_UNDER]->(:LookUp)-[:DENOTES]->(:Shade)
(:Verb)-[:FROM_ROOT]->(:Root)        (:Verb)-[:SEE_ALSO]->(:Verb)
```

Node	Meaning	Count
Sense	One atomic meaning, named by its lemma, e.g. गति (motion)	546
Shade	One distinct English gloss block under a Sense	577
Verb	A headword with its merged spelling and voice variants	1,525
LookUp	The original dataset label, kept for traceability	583
Root	Verbal root	1,298

In total there are 6,093 relationships, including 1,759 Verb–Shade links.

4. Building the relations

The data gives labels, not relations, so the pipeline (`pipeline/build_graph.py`) derives them:

1. **Split composite labels.** A label such as गतिहिंसयोः names two meanings at once. A curated table of compounds, qualifiers and aliases (`pipeline/lexicon.py`) splits it into गति + हिंसा.
2. **Divide the English gloss.** The gloss is assigned to each part: from a gloss the same meaning has elsewhere (only if another entry confirms it), otherwise by position (confidence 0.95), otherwise marked ambiguous (0.5).
3. **Merge near-duplicates.** Glosses under one Sense with word overlap of 80% or more become one Shade.
4. **Merge verb variants.** Forms listed together are merged, but prefixed verbs (प्राणिति / अपानिति) stay separate.
5. **Fill gaps.** Verbs without a label inherit meanings through See Also (confidence 0.9) or their root (0.5). Every link records its confidence and source.

5. Answering a query

1. **Read the input.** Statement text as is; .txt as UTF-8; PDF through PyMuPDF, with repair of broken Devanagari vowel signs and line wraps.
2. **Find the word.** Each occurrence is located, including inflected forms (चलन्ति → चलति).
3. **Collect context.** Other known verbs and meaning words in the passage are weighted by distance: same sentence 1.0, neighbouring sentence 0.5, elsewhere 0.1.
4. **Score each meaning.** Context words are compared with each candidate Shade through their English glosses (IDF-weighted cosine similarity). A meaning word from the dataset (e.g. कम्पे) adds a direct bonus.
5. **Choose.** The highest-scoring meaning is kept if the evidence is strong enough; otherwise the answer is marked as a best guess.
6. **Rank synonyms.** Verbs sharing that Shade are ordered by link confidence, then by how many other meanings they share with the word, then by fewest unrelated meanings. If none exists, verbs of the same Sense are offered as *related*.

When there is no answer, the app explains the reason: the word is not in the text, the verb is outside the dataset (which covers अ to न), it is a noun, the verb has no recorded meaning, or no other verb shares its meaning. It also suggests close spellings.

6. Worked example

Input statement: मुनिः नगरे क्ष्मायते । नरः कम्पे गेपते । राजा अपि एजति । Word: गेपते

Step	Result
Meanings of गेपते in the graph	गति (motion) and कम्प (trembling): two Shades
Context found	कम्पे, a meaning word in the same sentence (weight 1.0); क्ष्मायते and एजति, verbs in the neighbouring sentences (0.5), glossed *agitate, jerk, jolt*
Scores	कम्प 3.0, गति 0.09, so कम्प is selected
Answer	Best synonym केपते, with the other verbs of the same Shade listed as alternatives

With गतौ in place of कम्पे and motion verbs around it (test case T07), the same word resolves to गति instead.

7. Implementation

Part	Technology
Database	Neo4j 2025.12 Community, with uniqueness constraints and a full-text index on glosses
Pipeline and logic	Python 3.12, pandas, neo4j driver, PyMuPDF
Web app	Flask with one HTML page: choose Statement, Text file or PDF, enter the word, get the answer
Deployment	Windows: `start.bat`. Linux server: `setup_linux.sh` installs Java and Neo4j without root, `serve.sh` runs Neo4j and the app (gunicorn) for the network
Tests	17 unit tests, plus 20 generated paragraph cases in `test_data/`

8. Evaluation

Twenty paragraphs were generated from the dataset. They place the same verb in contexts for different meanings, add distracting cues one sentence away, use plural forms, and repeat a word twice with two meanings. All 22 occurrences received the intended meaning, and the PDF versions gave the same results. These cases contain the dataset's own meaning labels, so they confirm that the mechanism works rather than measure accuracy on real texts. No labelled Sanskrit corpus exists for that yet.

9. Limitations and next steps

- The verb list covers अ to न only. Common verbs such as वदति and भवति are absent.
- Only present-tense endings are recognised, not other tenses, participles or most sandhi.
- 146 verbs have no usable meaning and 120 Shades have no English gloss.
- Context is matched by words, not by sentence structure.
- Next steps: review `lexicon.py` with a Sanskritist, add a morphological analyser, build a labelled test set.